

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : **21**

Team Members :

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1. Problem Title:

Water Supply Management System

2. Problem Description:

Manual monitoring causes supply inefficiencies.

3. Identified Entities (Classes)

Class Name	Description
Consumer	Uses Water
Operator	Controls Supply
Admin	Oversees System
System	Tracks Usage

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Consumer	consumerId, name, address, waterUsage, billAmount	useWater(), viewUsage(), payBill(), reportLeakage()
Operator	operatorId, name, supplyZone, schedule	controlSupply(), updateSchedule(), monitorFlow(), stopSupply()
Admin	adminId, name, password, reports	addConsumer(), removeConsumer(), generateReport(), monitorSystem()
System	totalWaterSupply, usageData, leakageStatus, reportData	trackUsage(), detectLeakage(), generateConsumptionRep ort(), sendAlert()